

Adaptive Backstepping Path-Following Control of an Underactuated Quadcopter

Full Derivation with Zero Disturbances ($b = 0$)

Based on Xie, Cabecinhas, Cunha, Silvestre (2022)

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1 Notation

Throughout this document the following notation is used. $\mathbb{R}^n$ denotes the n -dimensional Euclidean space. For a vector $x \in \mathbb{R}^n$, its Euclidean norm is $\|x\| = \sqrt{x^\top x}$. For a matrix $\Theta \in \mathbb{R}^{n \times n}$, its Frobenius norm is $\|\Theta\|_F = (\text{trace}(\Theta^\top \Theta))^{1/2}$.

The map $S : \mathbb{R}^3 \rightarrow \mathbb{R}^{3 \times 3}$ yields the skew-symmetric matrix satisfying $S(x)y = x \times y$ for all $x, y \in \mathbb{R}^3$, i.e.,

$$S(x) = \begin{bmatrix} 0 & -x_3 & x_2 \\ x_3 & 0 & -x_1 \\ -x_2 & x_1 & 0 \end{bmatrix}, \quad x = [x_1, x_2, x_3]^\top.$$

Unit vectors are $e_1 = [1, 0, 0]^\top$, $e_2 = [0, 1, 0]^\top$, $e_3 = [0, 0, 1]^\top$. The hat $\hat{(\cdot)}$ denotes an estimate and $\tilde{(\cdot)} = (\cdot) - \hat{(\cdot)}$ its error. For a diagonal matrix $\Omega = \text{diag}(\Omega_1, \dots, \Omega_n)$, $\text{Diag}(\Omega) = [\Omega_1, \dots, \Omega_n]^\top$. The operator $\circ$ denotes $x \circ y = [x_1 y^\top, \dots, x_n y^\top]^\top$.

2 Vehicle Model

2.1 Coordinate Frames

Two coordinate frames are introduced:

- Inertial frame $\{I\}$: fixed to the ground.
- Body-fixed frame $\{B\}$: attached to the quadcopter's centre of mass.

The configuration of $\{B\}$ relative to $\{I\}$ is an element of the Special Euclidean group

$$(R, p) \in \text{SE}(3), \quad p \in \mathbb{R}^3, \quad R \in \text{SO}(3),$$

where p is the position of the centre of mass expressed in $\{I\}$, and R is the rotation matrix mapping vectors from $\{B\}$ to $\{I\}$.

2.2 Equations of Motion

The quadcopter is modelled as a rigid body. The kinematic equations are

$$\dot{p} = R v, \tag{1}$$

$$\dot{R} = R S(\omega), \tag{2}$$

and the dynamic equations are

$$\dot{v} = -S(\omega) v - m^{-1} T e_3 + g R^\top e_3, \quad (3)$$

$$\dot{\omega} = -J^{-1} S(\omega) J \omega + J^{-1} \tau, \quad (4)$$

where:

- $v \in \mathbb{R}^3$ — linear velocity expressed in $\{B\}$,
- $\omega \in \mathbb{R}^3$ — angular velocity expressed in $\{B\}$,
- $m \in \mathbb{R}$ — vehicle mass (unknown),
- $J \in \mathbb{R}^{3 \times 3}$ — moment of inertia matrix (unknown, assumed diagonal),
- g — gravitational acceleration,
- $T \in \mathbb{R}$ — total thrust force (control input),
- $\tau \in \mathbb{R}^3$ — torque vector (control input).

Remark 2.1: Zero disturbances

Throughout this document we set $\mathbf{b} = \mathbf{0}$, i.e. all external wind and aerodynamic disturbances are assumed absent. This simplifies the velocity dynamics (3) by removing the term $R^\top b$, and removes the need for a disturbance estimator. The mass m and moment of inertia J remain *unknown*.

2.3 Assumption on Unknown Parameters

Assumption 2.1. The vehicle mass m and moment of inertia J are constant and unknown, but satisfy

$$m \in [m_{\min}, m_{\max}], \quad \|J\|_F \in [J_{\min}, J_{\max}],$$

where $m_{\min}, m_{\max}, J_{\min}, J_{\max} > 0$ are known positive constants. Furthermore, J is diagonal due to vehicle symmetry: $J = \text{diag}(J_1, J_2, J_3)$.

3 Smooth Projection Function

The smooth projection function is used to keep parameter estimates inside known bounds, preventing singularities in the thrust law.

Definition 3.1 (Smooth projection). For given $s_{\min} < s_{\max}$ and arbitrarily small $\varepsilon > 0$, define

$$\pi(s) = \begin{cases} s_{\max} + \varepsilon(1 - e^{(s_{\max}-s)/\varepsilon}), & s > s_{\max}, \\ s, & s \in [s_{\min}, s_{\max}], \\ s_{\min} - \varepsilon(1 - e^{(s-s_{\min})/\varepsilon}), & s < s_{\min}. \end{cases}$$

Properties:

1. $\pi(s)$ is of class C^1 .
2. $\pi(s) \in [s_{\min} - \varepsilon, s_{\max} + \varepsilon]$ for all $s \in \mathbb{R}$.

To form a Lyapunov barrier for the estimation error, define $\tilde{s} = s - \hat{s}$ and

$$V(s, \tilde{s}) = \int_0^{\tilde{s}} (\pi(\varphi + \hat{s}) - s) d\varphi, \quad (5)$$

which is positive definite with respect to $\tilde{s}$ for $s \in [s_{\min}, s_{\max}]$. Its time derivative is

$$\dot{V}(s, \tilde{s}) = (\pi(\hat{s}) - s) \dot{\tilde{s}}. \quad (6)$$

Applied to the scalar $\eta = m^{-1}$ with bounds $\eta \in [m_{\max}^{-1}, m_{\min}^{-1}]$:

$$\pi(\hat{\eta}) \in [m_{\max}^{-1} - \varepsilon, m_{\min}^{-1} + \varepsilon].$$

Choosing $\varepsilon < m_{\max}^{-1}$ ensures $\pi(\hat{\eta}) > 0$ always, so the thrust law $T = \pi(\hat{\eta})^{-1}(\dots)$ is never singular.

4 Path-Following Problem

Definition 4.1 (Desired path). Let $p_d(\gamma) \in \mathbb{R}^3$ be a curve of class at least C^4 , parameterised by $\gamma \in \mathbb{R}$, with all partial derivatives with respect to γ bounded.

Control objective: Design a thrust law T , a torque law τ , and a timing law $\dot{\gamma}$ such that, for *any* initial position $p(0)$, the vehicle position $p(t)$ converges to an arbitrarily small neighbourhood of $p_d(\gamma(t))$.

4.1 Timing Law Design

Define the desired speed along the path as

$$\dot{\gamma}_d = v_d \psi(\|z_1\|), \quad (7)$$

where $v_d > 0$ is a constant desired speed, and $\psi : \mathbb{R}_{\geq 0} \rightarrow [0, 1]$ is the smooth bump-like function (C^∞):

$$\psi(s) = \begin{cases} 1, & 0 \leq |s| < \alpha_1, \\ \frac{1}{2} \tanh\left(\frac{\theta(s)}{\theta^2(s) - 1}\right) + \frac{1}{2}, & \alpha_1 \leq |s| \leq \alpha_2, \\ 0, & \alpha_2 < |s|, \end{cases}$$

with $\theta(s) = \frac{2s - \alpha_1 - \alpha_2}{\alpha_2 - \alpha_1}$ and design parameters $0 < \alpha_1 < \alpha_2$.

Interpretation:

- If $\|z_1\| > \alpha_2$: the path is stationary ($\dot{\gamma}_d = 0$), waiting for the drone.
- If $\|z_1\| \leq \alpha_1$: the path moves at full speed v_d .
- Intermediate: smooth blending.

5 Controller Derivation via Backstepping

The controller is derived by the backstepping technique. We build a composite Lyapunov function iteratively, introducing one error variable per step, until the physical inputs T and τ are obtained.

5.1 Step 1: Position Error z_1

Error definition

Define the position error in the inertial frame:

$$z_1 = p - p_d(\gamma). \quad (8)$$

Lyapunov candidate function V_1

To incorporate a saturation-type function for large errors, choose

$$V_1 = \sigma_{\max} \ln(\cosh(z_1))^{\top} \mathbf{1}_{3 \times 1} = \sigma_{\max} \sum_{i=1}^3 \ln \cosh(z_{1,i}), \quad (9)$$

where $\sigma_{\max} > 0$ is a design constant. Then:

$$\frac{\partial V_1}{\partial z_{1,i}} = \sigma_{\max} \tanh(z_{1,i}).$$

Define the saturating function $\sigma : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ component-wise:

$$\sigma(z_1) = \sigma_{\max} \tanh(z_1). \quad (10)$$

Time derivative of V_1

Differentiating (9) along trajectories:

$$\begin{aligned} \dot{V}_1 &= \sigma(z_1)^\top \dot{z}_1 \\ &= \sigma(z_1)^\top (\dot{p} - \dot{p}_d) \\ &\stackrel{(1)}{=} \sigma(z_1)^\top (Rv - \dot{p}_d). \end{aligned} \quad (11)$$

To create a negative-definite term, add and subtract $k_1 \sigma(z_1)^\top \sigma(z_1)$:

$$\dot{V}_1 = - \underbrace{k_1 \sigma(z_1)^\top \sigma(z_1)}_{W_1(z_1) \geq 0} + \sigma(z_1)^\top (Rv - \dot{p}_d + k_1 \sigma(z_1)), \quad (12)$$

where $k_1 > 0$ is a control gain and $W_1(z_1) = k_1 \|\sigma(z_1)\|^2$ is positive definite.

Handling underactuation — introduction of δ

The quadcopter is **underactuated**: thrust T acts only along the body z -axis (the e_3 direction). We cannot independently command all three components of Rv . A naive backstepping approach would require driving

$$Rv - \dot{p}_d + k_1 \sigma(z_1) \rightarrow 0_{3 \times 1},$$

which would require inverting R to extract v — this leads to singularities in the rotational subsystem. Instead, following [?], we drive this quantity to a small nonzero constant vector:

$$\delta = [0, 0, \delta]^\top \in \mathbb{R}^3, \quad \delta \neq 0. \quad (13)$$

Motivation: The term $R\delta$ stays bounded and, by choosing $\sigma_{\max} > \|\delta\|/k_1$, the resulting ultimate bound on z_1 can be made arbitrarily small.

Definition of z_2

Define the second error variable:

$$\boxed{z_2 = Rv - \dot{p}_d + k_1 \sigma(z_1) - R\delta.} \quad (14)$$

Then $Rv - \dot{p}_d + k_1\sigma(z_1) = z_2 + R\delta$, and (12) becomes:

$$\dot{V}_1 = -W_1(z_1) + \sigma(z_1)^\top (z_2 + R\delta). \quad (15)$$

5.2 Step 2: Velocity Error z_2 and Mass Estimation

Augmented Lyapunov candidate function V_2

Augment V_1 with $\frac{1}{2}z_2^\top z_2$:

$$V_2 = V_1 + \frac{1}{2}z_2^\top z_2. \quad (16)$$

Time derivative of z_2

We need $\dot{z}_2$. Differentiate (14):

$$\dot{z}_2 = \dot{R}v + R\dot{v} - \ddot{p}_d + k_1\dot{\sigma}(z_1) - \dot{R}\delta. \quad (17)$$

Substituting kinematic equation $\dot{R} = RS(\omega)$ and dynamic equation (3) (with $b = 0$):

$$\begin{aligned} \dot{z}_2 &= RS(\omega)v + R(-S(\omega)v - m^{-1}Te_3 + gR^\top e_3) - \ddot{p}_d + k_1\dot{\sigma}(z_1) - RS(\omega)\delta \\ &= \cancel{RS(\omega)v} - \cancel{RS(\omega)v} - Rm^{-1}Te_3 + gRR^\top e_3 - \ddot{p}_d + k_1\dot{\sigma}(z_1) - RS(\omega)\delta \\ &= -Rm^{-1}Te_3 + ge_3 - \ddot{p}_d + k_1\dot{\sigma}(z_1) - RS(\omega)\delta. \end{aligned} \quad (18)$$

Note on $\dot{\sigma}(z_1)$:

$$\dot{\sigma}(z_1) = \sigma_{\max} \text{diag}(\text{sech}^2(z_{1,i})) \dot{z}_1 = \sigma_{\max} \text{diag}(\text{sech}^2(z_{1,i})) (Rv - \dot{p}_d). \quad (19)$$

Time derivative of V_2

$$\begin{aligned} \dot{V}_2 &= \dot{V}_1 + z_2^\top \dot{z}_2 \\ &= -W_1(z_1) + \sigma(z_1)^\top (z_2 + R\delta) + z_2^\top (-Rm^{-1}Te_3 + ge_3 - \ddot{p}_d + k_1\dot{\sigma}(z_1) - RS(\omega)\delta). \end{aligned} \quad (20)$$

Introduce $\eta = m^{-1}$ and write the term in $z_2^\top(\dots)$ more compactly by defining:

$$\zeta = R^\top (ge_3 - \ddot{p}_d + k_1\dot{\sigma}(z_1)), \quad \text{so that } ge_3 - \ddot{p}_d + k_1\dot{\sigma}(z_1) = R\zeta.$$

Then:

$$\begin{aligned}\dot{V}_2 &= -W_1(z_1) + \sigma(z_1)^\top (z_2 + R\delta) \\ &\quad + z_2^\top R(-\eta T e_3 - S(\omega) \delta + \zeta). \end{aligned} \quad (21)$$

Add and subtract $k_2 z_2^\top \sigma(z_2)$ to create the next positive-definite term:

$$\begin{aligned}\dot{V}_2 &= -W_2(z_1, z_2) + \sigma(z_1)^\top (z_2 + R\delta) \\ &\quad + z_2^\top R(-\eta T e_3 - S(\omega) \delta + \zeta + k_2 \sigma(z_2)), \end{aligned} \quad (22)$$

where

$$W_2(z_1, z_2) = W_1(z_1) + k_2 z_2^\top \sigma(z_2), \quad k_2 > 0.$$

Handling unknown $\eta = m^{-1}$

Since η is unknown, we introduce an estimate $\hat{\eta}$ and an estimation error $\tilde{\eta} = \eta - \hat{\eta}$. Adding the Lyapunov term for the estimation error using (5):

$$V_{2b} = V_2 + \lambda_\eta^{-1} V(\eta, \tilde{\eta}), \quad (23)$$

where $\lambda_\eta > 0$.

Using (6), the time derivative is:

$$\dot{V}_{2b} = \dot{V}_2 + \lambda_\eta^{-1} (\pi(\hat{\eta}) - \eta) \dot{\hat{\eta}}. \quad (24)$$

Now rewrite the $z_2^\top R(-\eta T e_3 + \dots)$ term. Replace η by $\pi(\hat{\eta}) + (\eta - \pi(\hat{\eta})) \approx \pi(\hat{\eta}) - \tilde{\eta}$ (since $\pi(\hat{\eta}) \approx \hat{\eta}$ inside the bounds):

$$\begin{aligned}\dot{V}_{2b} &= -W_2(z_1, z_2) + \sigma(z_1)^\top (z_2 + R\delta) \\ &\quad + z_2^\top R(-\pi(\hat{\eta}) T e_3 - S(\omega) \delta + \zeta + k_2 \sigma(z_2)) \\ &\quad + \tilde{\eta} \left(z_2^\top R e_3 T + \lambda_\eta^{-1} \dot{\hat{\eta}} \right). \end{aligned} \quad (25)$$

To zero out the last $\tilde{\eta}$ term, choose the **mass estimator update law**:

$$\boxed{\dot{\hat{\eta}} = -\lambda_\eta z_2^\top R^\top e_3 T.} \quad (26)$$

Substituting (26) into (25):

$$\begin{aligned}\dot{V}_{2b} &= -W_2(z_1, z_2) + \sigma(z_1)^\top (z_2 + R\delta) \\ &\quad + z_2^\top R(-\pi(\hat{\eta}) T e_3 - S(\omega) \delta + \zeta + k_2 \sigma(z_2)). \end{aligned} \quad (27)$$

Decomposition via Π_{12} and thrust design

Define the projection matrix

$$\Pi_{12} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \in \mathbb{R}^{2 \times 3},$$

so that $e_3 e_3^\top = I_3 - \Pi_{12}^\top \Pi_{12}$, and write:

$$z_2^\top R(-\pi(\hat{\eta}) T e_3 - S(\omega) \delta + \zeta + k_2 \sigma(z_2)) = z_2^\top R[e_3 e_3^\top + (I_3 - e_3 e_3^\top)](-\pi(\hat{\eta}) T e_3 - S(\omega) \delta + \zeta + k_2 \sigma(z_2))$$

Split into the e_3 -component (cancelled by T) and the orthogonal complement (handled by ω):

$$\begin{aligned} \dot{V}_{2b} = & -W_2(z_1, z_2) + \sigma(z_1)^\top (z_2 + R\delta) \\ & + z_2^\top R e_3 e_3^\top (-\pi(\hat{\eta}) T e_3 - S(\omega) \delta + \zeta + k_2 \sigma(z_2)) \\ & + z_2^\top R (I_3 - e_3 e_3^\top) (-S(\omega) \delta + \zeta + k_2 \sigma(z_2)). \end{aligned} \quad (28)$$

To cancel the $e_3 e_3^\top$ term, design the **thrust force**:

$$T = \pi(\hat{\eta})^{-1} e_3^\top (-S(\omega) \delta + \zeta + k_2 \sigma(z_2)), \quad (29)$$

where $\pi(\hat{\eta}) > 0$ is guaranteed by the projection function.

After substituting (29) the $e_3 e_3^\top$ term vanishes, leaving:

$$\dot{V}_{2b} = -W_2(z_1, z_2) + \sigma(z_1)^\top (z_2 + R\delta) + (\Pi_{12} R^\top z_2)^\top (\delta_n \omega + \Pi_{12} \zeta_{\text{aug}}), \quad (30)$$

where

$$\delta_n = \begin{bmatrix} 0 & -\delta & 0 \\ \delta & 0 & 0 \end{bmatrix}, \quad \zeta_{\text{aug}} = \zeta + k_2 \sigma(z_2).$$

Virtual desired angular velocity ω_d

To cancel the last term, choose the virtual desired angular velocity:

$$\omega_d = -\delta_m \zeta_{\text{aug}}, \quad \delta_m = \delta_n^\top (\delta_n \delta_n^\top)^{-1} \Pi_{12}. \quad (31)$$

Note: $\det(\delta_n \delta_n^\top) = \delta^4 \neq 0$ (since $\delta \neq 0$), so δ_m is always well defined. With $\omega = \omega_d + z_3$ (to be defined), substituting back:

$$\delta_n \omega + \Pi_{12} \zeta_{\text{aug}} = \delta_n (\omega_d + z_3) + \Pi_{12} \zeta_{\text{aug}} = \cancel{\delta_n \omega_d + \Pi_{12} \zeta_{\text{aug}}} + \delta_n z_3 = \delta_n z_3.$$

Hence (30) reduces to:

$$\dot{V}_{2b} = -W_2(z_1, z_2) + \sigma(z_1)^\top (z_2 + R\delta) + (\Pi_{12}R^\top z_2)^\top \delta_n z_3. \quad (32)$$

5.3 Step 3: Angular Velocity Error z_3 and Inertia Estimation

Error definition

Define the angular velocity error:

$$z_3 = \omega - \omega_d. \quad (33)$$

Augmented Lyapunov candidate function V_3

$$V_3 = V_{2b} + \frac{\beta}{2} z_3^\top J z_3, \quad \beta > 0. \quad (34)$$

Time derivative of z_3

$$\dot{z}_3 = \dot{\omega} - \dot{\omega}_d = -J^{-1}S(\omega) J\omega + J^{-1}\tau - \dot{\omega}_d.$$

Time derivative of V_3

$$\begin{aligned} \dot{V}_3 &= \dot{V}_{2b} + \beta z_3^\top J \dot{z}_3 \\ &= -W_2(z_1, z_2) + \sigma(z_1)^\top (z_2 + R\delta) + (\Pi_{12}R^\top z_2)^\top \delta_n z_3 \\ &\quad + \beta z_3^\top J (-J^{-1}S(\omega) J\omega + J^{-1}\tau - \dot{\omega}_d) \\ &= -W_2(z_1, z_2) + \sigma(z_1)^\top (z_2 + R\delta) + (\Pi_{12}R^\top z_2)^\top \delta_n z_3 \\ &\quad + \beta z_3^\top (-S(\omega) J\omega + \tau + J(-\dot{\omega}_d)). \end{aligned} \quad (35)$$

Add and subtract $k_3 z_3^\top \sigma(z_3)$:

$$\begin{aligned} \dot{V}_3 &= -W_3(z_1, z_2, z_3) + \sigma(z_1)^\top (z_2 + R\delta) \\ &\quad + \beta z_3^\top [\tau - S(\omega) J\omega - J\dot{\omega}_d + \beta^{-1}(\delta_n^\top \Pi_{12}R^\top z_2 + k_3 \sigma(z_3))], \end{aligned} \quad (36)$$

where

$$W_3(z_1, z_2, z_3) = W_2(z_1, z_2) + k_3 z_3^\top \sigma(z_3), \quad k_3 > 0.$$

Factoring out the unknown parameters

The term $-S(\omega) J\omega - J\dot{\omega}_d$ contains the unknown J . We factor this as $\mathcal{W}w$ where $\mathcal{W} \in \mathbb{R}^{3 \times p}$ is a known, state-dependent matrix and $w \in \mathbb{R}^p$ collects all unknown constant

parameters:

$$-S(\omega) J\omega - J\dot{\omega}_d = \mathcal{W}(\omega, \dot{\omega}_d) w, \quad (37)$$

with

$$w = \text{Diag}(J) = [J_1, J_2, J_3]^\top. \quad (38)$$

Explicit construction of $\mathcal{W}$:

With $J = \text{diag}(J_1, J_2, J_3)$, $S(\omega) J\omega$ expands as:

$$S(\omega) J\omega = \begin{bmatrix} \omega_2\omega_3(J_3 - J_2) \\ \omega_1\omega_3(J_1 - J_3) \\ \omega_1\omega_2(J_2 - J_1) \end{bmatrix} = \underbrace{\begin{bmatrix} -\omega_2\omega_3 & \omega_2\omega_3 & 0 \\ \omega_1\omega_3 & 0 & -\omega_1\omega_3 \\ 0 & -\omega_1\omega_2 & \omega_1\omega_2 \end{bmatrix}}_{\mathcal{W}_1(\omega)} \begin{bmatrix} J_1 \\ J_2 \\ J_3 \end{bmatrix}.$$

Similarly, $J\dot{\omega}_d = \text{diag}(\dot{\omega}_d) \cdot [J_1, J_2, J_3]^\top$, so $-J\dot{\omega}_d = -\text{diag}(\dot{\omega}_d) w$.

Therefore:

$$\mathcal{W} = \mathcal{W}_1(\omega) - \text{diag}(\dot{\omega}_d), \quad w = [J_1, J_2, J_3]^\top.$$

Augmented Lyapunov function with inertia estimation error

Introduce an estimate $\hat{w}$ of w , with error $\tilde{w} = w - \hat{w}$. Augment:

$$V_{3b} = V_3 + \frac{1}{2}\tilde{w}^\top \Lambda_w^{-1} \tilde{w}, \quad (39)$$

where $\Lambda_w \in \mathbb{R}^{3 \times 3}$ is a positive-definite diagonal gain matrix.

Time derivative of V_{3b}

$$\begin{aligned} \dot{V}_{3b} &= -W_3(z_1, z_2, z_3) + \sigma(z_1)^\top (z_2 + R\delta) \\ &\quad + \beta z_3^\top [\tau + \mathcal{W}\hat{w} + \beta^{-1}(\delta_n^\top \Pi_{12} R^\top z_2 + k_3 \sigma(z_3))] \\ &\quad + \beta z_3^\top \mathcal{W}\tilde{w} + \tilde{w}^\top \Lambda_w^{-1} \dot{\tilde{w}} - \beta z_3^\top \mathcal{W}\tilde{w} \\ &= -W_3 + \sigma(z_1)^\top (z_2 + R\delta) \\ &\quad + \beta z_3^\top [\tau + \mathcal{W}\hat{w} + \beta^{-1}(\delta_n^\top \Pi_{12} R^\top z_2 + k_3 \sigma(z_3))] \\ &\quad + \tilde{w}^\top \left(-\beta \mathcal{W}^\top z_3 + \Lambda_w^{-1} \dot{\tilde{w}} \right). \end{aligned} \quad (40)$$

Torque design and inertia estimator

To cancel the second and third bracketed terms in (40):

Design the **torque input**:

$$\tau = -\mathcal{W}\hat{w} - \beta^{-1}(\delta_n^\top \Pi_{12} R^\top z_2 + k_3 \sigma(z_3)), \quad (41)$$

and the **inertia estimator update law**:

$$\dot{\hat{w}} = \beta \Lambda_w \mathcal{W}^\top z_3. \quad (42)$$

Substituting (41) and (42) into (40), all non-negative-definite terms cancel:

$$\dot{V}_{3b} = -W_3(z_1, z_2, z_3) + \sigma(z_1)^\top (z_2 + R\delta). \quad (43)$$

5.4 Step 4: Path Timing Law

Define the along-path speed tracking error:

$$z_4 = \dot{\gamma} - \dot{\gamma}_d. \quad (44)$$

Choose the Lyapunov function $V_4 = \frac{1}{2}z_4^2$. Its time derivative:

$$\dot{V}_4 = z_4 \dot{z}_4 = z_4(\ddot{\gamma} - \ddot{\gamma}_d + k_4 z_4 - k_4 z_4) = -k_4 z_4^2 + z_4(\ddot{\gamma} - \ddot{\gamma}_d + k_4 z_4). \quad (45)$$

Design the **timing law**:

$$\ddot{\gamma} = \ddot{\gamma}_d - k_4 z_4, \quad (46)$$

yielding:

$$\dot{V}_4 = -k_4 z_4^2 \leq 0. \quad (47)$$

6 Summary of the Complete Control Law

The complete adaptive backstepping controller consists of the following laws:

Thrust force

$$T = \pi(\hat{\eta})^{-1} e_3^\top (-S(\omega) \delta + \zeta + k_2 \sigma(z_2)),$$

where $\zeta = R^\top (g e_3 - \ddot{p}_d + k_1 \dot{\sigma}(z_1))$.

Torque

$$\tau = -\mathcal{W}\hat{w} - \beta^{-1}(\delta_n^\top \Pi_{12} R^\top z_2 + k_3 \sigma(z_3)).$$

Adaptive estimators

$$\begin{aligned}\dot{\hat{\eta}} &= -\lambda_{\eta} z_2^{\top} R^{\top} e_3 T, \quad (\text{mass estimate, } \hat{\eta} = \hat{m}^{-1}) \\ \dot{\hat{w}} &= \beta \Lambda_w \mathcal{W}^{\top} z_3. \quad (\text{inertia estimate, } w = [J_1, J_2, J_3]^{\top})\end{aligned}$$

Timing law

$$\ddot{\gamma} = \ddot{\gamma}_d - k_4 z_4, \quad \dot{\gamma}_d = v_d \psi(\|z_1\|).$$

where all error variables are:

$$z_1 = p - p_d, \quad z_2 = Rv - \dot{p}_d + k_1 \sigma(z_1) - R\delta, \quad z_3 = \omega - \omega_d, \quad z_4 = \dot{\gamma} - \dot{\gamma}_d.$$

7 Stability Analysis

Theorem 7.1: Global Uniform Ultimate Boundedness

Let the quadcopter kinematics and dynamics be described by (1)–(4) with $b = 0$, and let $p_d(\gamma) \in C^4$ be the desired path with bounded partial derivatives. Under Assumption 3.1, with the control laws (29), (41), estimator laws (26), (42), timing law (46), and projection (5), where $\sigma_{\max}$ satisfies $\sigma_{\max} > \|\delta\|/k_1$, the following holds for any initial condition:

1. The position error z_1 is globally uniformly ultimately bounded (GUUB) with $\|z_1(t)\| \rightarrow \mathcal{B}_1 := \sigma^{-1}(\|\delta\|/k_1)$.
2. The errors z_2, z_3, z_4 converge to zero as $t \rightarrow \infty$.
3. All parameter estimates $\hat{\eta}, \hat{w}$ remain bounded for all $t \geq 0$.

Proof. The proof proceeds in three parts.

Part I: Boundedness and convergence of (z_2, z_3) .

Consider the Lyapunov function

$$V_n = \frac{1}{2} z_2^{\top} z_2 + \frac{\beta}{2} z_3^{\top} J z_3 + \frac{1}{2} \tilde{w}^{\top} \Lambda_w^{-1} \tilde{w} + \lambda_{\eta}^{-1} V(\eta, \tilde{\eta}), \quad (48)$$

which is positive definite and radially unbounded in $(z_2, z_3, \tilde{w}, \tilde{\eta})$.

From (43), and noting that $\sigma(z_1)^{\top}(z_2 + R\delta)$ is absent from V_n (it arises from the V_1

part), the time derivative of V_n is:

$$\dot{V}_n = z_2^\top \dot{z}_2 + \beta z_3^\top J \dot{z}_3 - \tilde{w}^\top \Lambda_w^{-1} \dot{\tilde{w}} + \lambda_\eta^{-1} (\pi(\hat{\eta}) - \eta) \dot{\hat{\eta}}. \quad (49)$$

Expanding each part using (18), (4), (26), (42):

$$\begin{aligned} z_2^\top \dot{z}_2 &= z_2^\top R(-\eta T e_3 - S(\omega) \delta + \zeta + k_2 \sigma(z_2)) \\ &= z_2^\top R(-\pi(\hat{\eta}) T e_3 + \tilde{\eta} T e_3 - S(\omega) \delta + \zeta + k_2 \sigma(z_2)) \\ &\stackrel{(29)}{=} z_2^\top R(\tilde{\eta} T e_3 - (I - e_3 e_3^\top)(S(\omega) \delta - \zeta - k_2 \sigma(z_2))). \end{aligned}$$

After applying thrust law (29) and rearranging via virtual control ω_d :

$$z_2^\top \dot{z}_2 = -k_2 z_2^\top \sigma(z_2) + (\Pi_{12} R^\top z_2)^\top \delta_n z_3 + \tilde{\eta} (z_2^\top R e_3 T).$$

Similarly:

$$\beta z_3^\top J \dot{z}_3 = \beta z_3^\top (\tau + \mathcal{W} w - J \dot{\omega}_d - S(\omega) J \omega) = -k_3 z_3^\top \sigma(z_3) - (\Pi_{12} R^\top z_2)^\top \delta_n z_3 + \beta z_3^\top \mathcal{W} \tilde{w}.$$

The cross-term $(\Pi_{12} R^\top z_2)^\top \delta_n z_3$ cancels between the two. Using update laws:

$$-\tilde{w}^\top \Lambda_w^{-1} \dot{\tilde{w}} = -\beta \tilde{w}^\top \mathcal{W}^\top z_3, \quad \lambda_\eta^{-1} (\pi(\hat{\eta}) - \eta) \dot{\hat{\eta}} = -\tilde{\eta} (z_2^\top R e_3 T).$$

Collecting all terms:

$$\dot{V}_n = -k_2 z_2^\top \sigma(z_2) - k_3 z_3^\top \sigma(z_3). \quad (50)$$

Since $\sigma(x) = \sigma_{\max} \tanh(x)$ has the same sign as x component-wise, $z_i^\top \sigma(z_i) > 0$ for $z_i \neq 0$. Hence $\dot{V}_n \leq 0$, so V_n is non-increasing. This immediately implies:

$$z_2, z_3, \tilde{w}, \tilde{\eta} \text{ are all bounded for all } t \geq 0. \quad (51)$$

In particular, $\|\tilde{\eta}\| \leq c_1$ and $\|\tilde{w}\| \leq c_2$ for some finite constants.

Barbalat's Lemma argument:

We now show $z_2 \rightarrow 0$ and $z_3 \rightarrow 0$.

Since the system states (p, R, v, ω) evolve according to bounded dynamics (Assumption 3.1 bounds m, J , and the control inputs are bounded functions of the states), all state trajectories are bounded. Therefore $\ddot{V}_n$ is bounded, which implies $\dot{V}_n$ is uniformly continuous.

Since $V_n \geq 0$ and $\dot{V}_n \leq 0$, V_n is bounded from below and non-increasing, hence

$\lim_{t \rightarrow \infty} V_n(t)$ exists. Therefore $\int_0^\infty (-\dot{V}_n) dt < \infty$, i.e.,

$$\int_0^\infty (k_2 z_2^\top \sigma(z_2) + k_3 z_3^\top \sigma(z_3)) dt < \infty.$$

By **Barbalat's Lemma**¹, since $k_2 z_2^\top \sigma(z_2) + k_3 z_3^\top \sigma(z_3)$ is uniformly continuous and integrable:

$$k_2 z_2^\top \sigma(z_2) + k_3 z_3^\top \sigma(z_3) \rightarrow 0 \quad \text{as } t \rightarrow \infty. \quad (52)$$

Since $x^\top \sigma(x) = \sigma_{\max} \sum_i x_i \tanh(x_i) = 0$ iff $x = 0$, we conclude:

$$z_2(t) \rightarrow 0, \quad z_3(t) \rightarrow 0 \quad \text{as } t \rightarrow \infty. \quad (53)$$

Part II: GUUB of z_1 .

Return to (15):

$$\dot{V}_1 = -W_1(z_1) + \sigma(z_1)^\top (z_2 + R\delta).$$

Bound the cross-term using $\|R\delta\| = \|\delta\|$:

$$\begin{aligned} \dot{V}_1 &\leq -k_1 \|\sigma(z_1)\|^2 + \|\sigma(z_1)\|(\|\delta\| + \|z_2\|) \\ &= -\|\sigma(z_1)\|(k_1 \|\sigma(z_1)\| - \|\delta\| - \|z_2\|). \end{aligned} \quad (54)$$

This is strictly negative whenever

$$k_1 \|\sigma(z_1)\| > \|\delta\| + \|z_2\|,$$

i.e., whenever $\|\sigma(z_1)\| > \frac{\|\delta\| + \|z_2\|}{k_1}$.

Since $z_2(t) \rightarrow 0$ by Part I, for any $\varepsilon > 0$ there exists T_ε such that $\|z_2(t)\| < \varepsilon$ for all $t \geq T_\varepsilon$. For $t \geq T_\varepsilon$:

$$\dot{V}_1 < 0 \quad \text{whenever} \quad \|\sigma(z_1)\| > \frac{\|\delta\| + \varepsilon}{k_1}.$$

Taking $\varepsilon \rightarrow 0$, the ultimate bound on $\sigma(\|z_1\|)$ is $\|\delta\|/k_1$. Applying σ^{-1} (which exists since $\sigma_{\max} > \|\delta\|/k_1$ by assumption):

$$\limsup_{t \rightarrow \infty} \|z_1(t)\| \leq \sigma^{-1}\left(\frac{\|\delta\|}{k_1}\right) = \operatorname{atanh}\left(\frac{\|\delta\|}{k_1 \sigma_{\max}}\right). \quad (55)$$

Global uniform ultimate boundedness follows because V_1 is radially unbounded in z_1 (since $\ln \cosh(z_{1,i}) \rightarrow \infty$ as $|z_{1,i}| \rightarrow \infty$), so large initial errors only increase $V_1(0)$ and the bound (55) holds regardless of initial conditions.

Part III: Convergence of z_4 .

¹If $f : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}$ is uniformly continuous and $\int_0^\infty f(t) dt$ exists and is finite, then $f(t) \rightarrow 0$.

From (47):

$$\dot{V}_4 = -k_4 z_4^2 = -2k_4 V_4,$$

which gives directly:

$$z_4(t) = z_4(0) e^{-k_4 t} \rightarrow 0 \quad \text{exponentially.} \quad (56)$$

Combining Parts I–III completes the proof. $\square$ $\square$

8 Discussion of the Bounds and Design Parameters

Remark 8.1: Tradeoff between accuracy and actuation magnitude

From (55), the ultimate position error bound decreases with:

- Larger k_1 (position gain),
- Smaller $\|\delta\|$ (offset magnitude).

However, from the thrust law (29), $T \propto k_1$, and from the torque law (41), $\tau \propto \delta^{-4}$ (via $\delta_m \propto \delta^{-4}$). Therefore making $\delta \rightarrow 0$ causes the torque to blow up. A design tradeoff is necessary.

Remark 8.2: Comparison with GAS

The result is GUUB rather than globally asymptotically stable (GAS) because the offset $\delta \neq 0$ introduces a persistent bounded “disturbance” into the z_1 subsystem. If all disturbances were zero AND $\delta = 0$ were achievable (which would require the vehicle to be fully actuated), the controller would give GAS. The price of handling underactuation is a residual bounded error.

Remark 8.3: Role of the projection function

The smooth projection $\pi(\hat{\eta})$ ensures $\pi(\hat{\eta}) > 0$ always, so the thrust $T = \pi(\hat{\eta})^{-1}(\dots)$ is never singular. Without projection, $\hat{\eta}$ could drift negative, making T change sign — physically impossible and control-theoretically catastrophic. The projection also enables the clean Lyapunov analysis via the barrier function $V(\eta, \tilde{\eta})$ in (5).

9 Numerical Parameters (Simulation Reference)

Parameter	Value	Role
k_1	4	Position gain
k_2	4	Velocity gain
k_3	13	Angular velocity gain
k_4	1	Path speed gain
$\lambda_m (= \lambda_\eta)$	0.3	Mass estimator gain
δ	0.1	Underactuation offset
$\sigma_{\max}$	1.5	Saturation level
α_1	0.5 m	Path timing inner threshold
α_2	1.5 m	Path timing outer threshold
v_d	design-specific	Desired path speed
Ultimate bound	$\sigma^{-1}(\delta/k_1) \approx 0.0167$ m	Theoretical max steady error

Appendix: Key Mathematical Identities

A.1 Skew-symmetric matrix identity:

$$S(x)y = -S(y)x = x \times y, \quad S(x)^\top = -S(x).$$

A.2 Rotation matrix property:

$$R^\top R = RR^\top = I_3, \quad \det(R) = 1.$$

A.3 Saturation function derivative:

$$\frac{d}{dt} \sigma_{\max} \tanh(z_1) = \sigma_{\max} \operatorname{diag}(\operatorname{sech}^2(z_{1,i})) \dot{z}_1.$$

A.4 Matrix δ_n and its inverse:

$$\delta_n = \begin{bmatrix} 0 & -\delta & 0 \\ \delta & 0 & 0 \end{bmatrix}, \quad \delta_n \delta_n^\top = \delta^2 I_2, \quad (\delta_n \delta_n^\top)^{-1} = \delta^{-2} I_2.$$

A.5 Barbalat's Lemma:

If $f : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}$ is uniformly continuous and $\int_0^\infty f(t) dt < \infty$, then $f(t) \rightarrow 0$ as $t \rightarrow \infty$.

A.6 Globally Uniform Ultimate Boundedness (GUUB):

A signal $x(t)$ is GUUB with bound r if there exists $T(x(0), r)$ such that $\|x(t)\| \leq r$ for all $t \geq T$, independent of the initial state.